arXiv:2211.11905 Fig.3–Fig.7 复现对比汇总

本 PDF 汇总 Benchmark/260928 COHENRENT-NSI/native_overlap_comparison 的输出。结果性质为 PARTIAL_NATIVE_APPROXIMATE_COMPARISON：使用迁移后的 CEvNS-pyEFT Flavor Fast kernel 与 legacy approximate statistics，不是论文 paper-matched likelihood 复现。
Fig.3–Fig.6 生成了 native approximate NSI 重绘和论文页渲染的并排对比。Fig.7 是 NGI scalar/vector/tensor，当前没有用 NSI 结果伪造复现，已标为 NGI_MISSING / NOT_COMPARABLE。
paper_sha256: 071225342b004f4e97001d6c200066b216f0af1af226f4b4aafa43512304a2b1
git_commit: 880dabab9a54911caa247c7dda12781e8c970123
grid_n: 101; threshold_2d: 4.605170185988092

关键限制

1. CsI 使用当前 CsI2021 legacy approximate likelihood，不是论文 9 PE x 11 timing Poisson likelihood。
2. Ar 使用当前 Ar2020 legacy approximate likelihood，不是论文 LAr timing-energy Gaussian + shape nuisance likelihood。
3. 正式 paper joint 不加入 CsI2017；本对比也未加入 CsI2017。
4. 论文原图为整页渲染，尚非人工精裁后的图像区域。

Panel 状态表

以下表格来自 panel_status.csv；Fig.7 明确标为不可比。

figure	panel	x	y	status
Fig3	top	eps_ee_dV	eps_ee_uV	NATIVE_APPROXIMATE_GENERATED
Fig3	bottom	eps_mumu_dV	eps_mumu_uV	NATIVE_APPROXIMATE_GENERATED
Fig4	top	eps_ee_uV	eps_mumu_uV	NATIVE_APPROXIMATE_GENERATED
Fig4	bottom	eps_ee_dV	eps_mumu_dV	NATIVE_APPROXIMATE_GENERATED
Fig5	top	eps_ee_dV	eps_etau_dV	NATIVE_APPROXIMATE_GENERATED
Fig5	bottom	eps_mumu_dV	eps_mutau_dV	NATIVE_APPROXIMATE_GENERATED
Fig6	top	eps_ee_dV	eps_emu_dV	NATIVE_APPROXIMATE_GENERATED
Fig6	bottom	eps_mumu_dV	eps_emu_dV	NATIVE_APPROXIMATE_GENERATED
Fig7	all	NGI	NGI	NGI_MISSING_NOT_COMPARABLE

Fig03 native approx comparison

左侧为论文 PDF 对应页渲染；右侧为当前 CEvNS-pyEFT native approximate 重绘。该页用于视觉对照和实现差异定位，不表示已完成 paper-matched 统计复现。

Paper original page render. Fig. 3 area must be manually cropped/audited

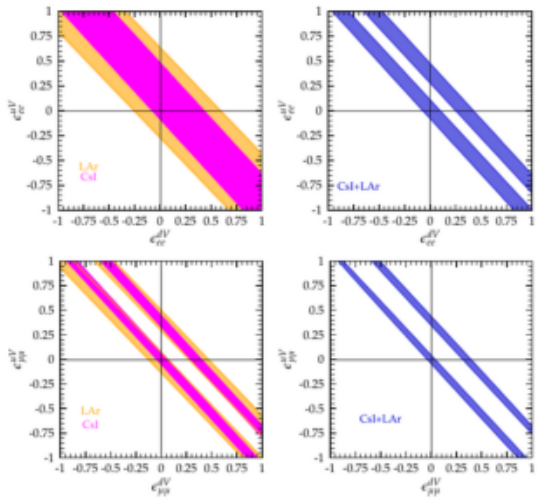


FIG. 3: 90% C.L. (2 d.o.f.) allowed regions on flavor-preserving neutrino NSI from the CsI (magenta), LAr (orange) and CsI + LAr (blue) analyses, allowing two NSI parameters at a time. The analysis of CsI data includes only CEνNS interactions.

In addition, our result is consistent with the analysis of CsI (2017) data presented in [64].

Finally, in Figs. 5 and 6 we display the allowed regions for one flavor-preserving NSI versus one flavor-changing NSI parameter for various combinations, $(\epsilon_{ee}^{dV}, \epsilon_{e\tau}^{dV})$, $(\epsilon_{\mu\mu}^{dV}, \epsilon_{\mu\tau}^{dV})$, and $(\epsilon_{ee}^{dV}, \epsilon_{e\mu}^{dV})$ $(\epsilon_{\mu\mu}^{dV}, \epsilon_{e\mu}^{dV})$, respectively. As previously, the results show the constraints from the CsI or LAr measurements (left panels), as well as those resulting from the combined CsI+LAr analysis (right panels), leading to the same general conclusions as discussed above. The implications of the enlarged CsI (2021) dataset can be seen, for instance, in the $(\epsilon_{ee}^{dV}, \epsilon_{e\tau}^{dV})$ plane, for which the allowed region is reduced by $\sim 40\%$ in comparison with the results obtained using the former CsI (2017) data [96]. To summarise, the 1σ limits obtained from the combined CsI+LAr analysis on the single

taken by the COHERENT Collaboration. In contrast, here we have considered 2 d.o.f., which leads to a single band.

CEvNS-pyEFT native approximate reproduction

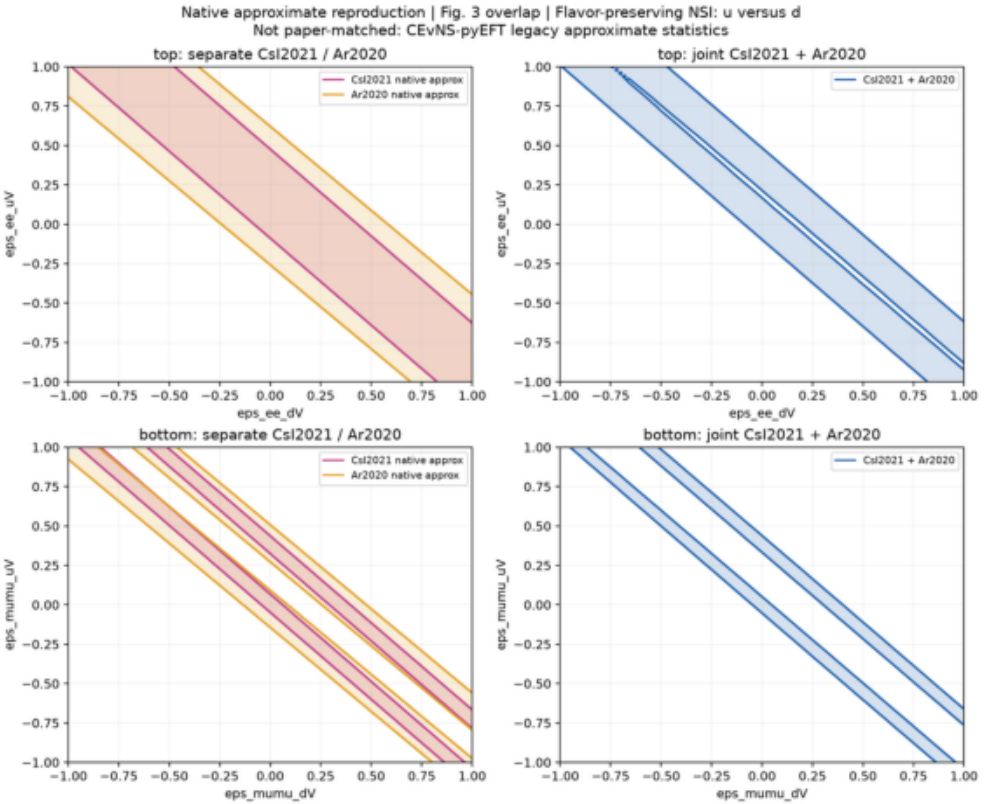


Fig04 native approx comparison

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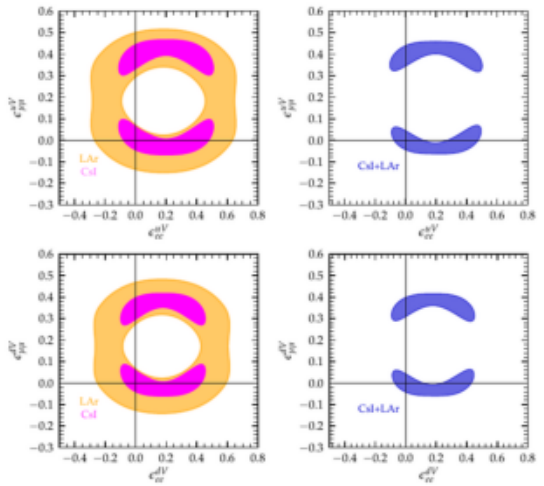


FIG. 4: Same as Fig. 3 but in the $(\epsilon_{ee}^{qV}, \epsilon_{\mu\mu}^{qV})$ planes.

nonuniversal NSI parameters (setting all the others to zero) read

$$\begin{aligned} \epsilon_{ee}^{dV} &= [-0.027, 0.048] \cup [0.30, 0.39], \\ \epsilon_{ee}^{uV} &= [-0.024, 0.045] \cup [0.34, 0.43], \\ \epsilon_{\mu\mu}^{dV} &= [-0.012, 0.016] \cup [0.33, 0.37], \\ \epsilon_{\mu\mu}^{uV} &= [-0.002, 0.001] \cup [0.37, 0.41]. \end{aligned} \tag{43}$$

Similarly, the 1σ limits on flavor-changing NSI parameters are

$$\begin{aligned} \epsilon_{e\mu}^{dV} &= [-0.071, 0.071], \quad \epsilon_{e\mu}^{uV} = [-0.081, 0.081], \\ \epsilon_{e\tau}^{dV} &= [-0.12, 0.12], \quad \epsilon_{e\tau}^{uV} = [-0.13, 0.13], \\ \epsilon_{\mu\tau}^{dV} &= [-0.087, 0.087], \quad \epsilon_{\mu\tau}^{uV} = [-0.098, 0.098]. \end{aligned} \tag{44}$$

CEvNS-pyEFT native approximate reproduction

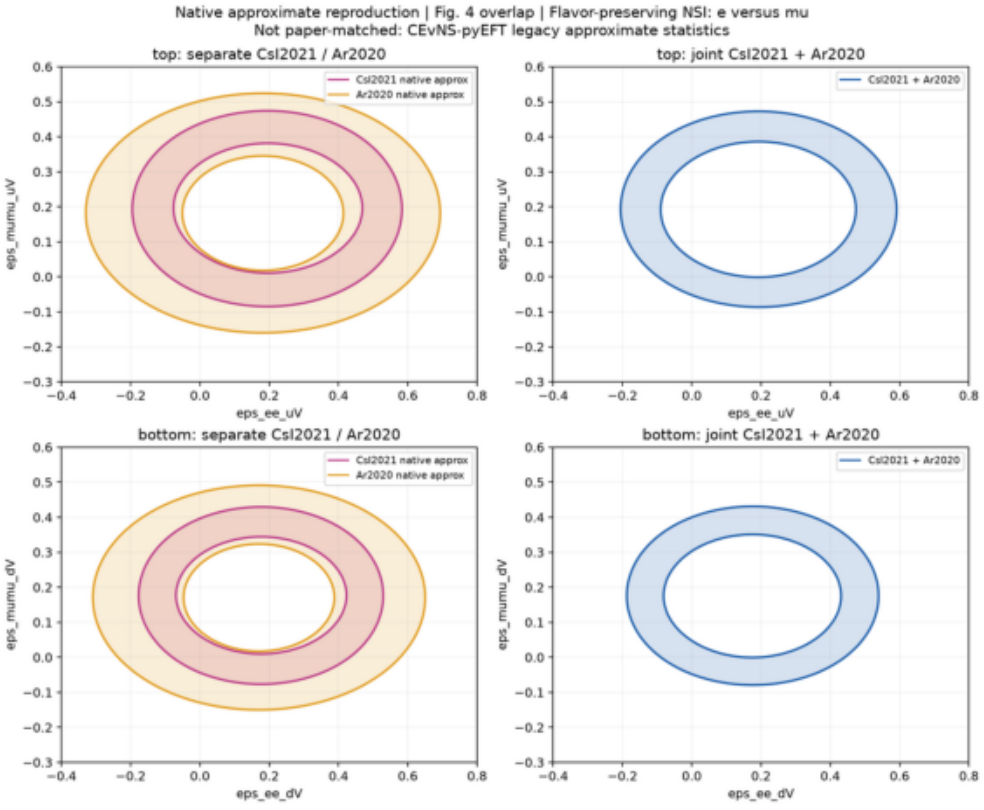


Fig05 native approx comparison

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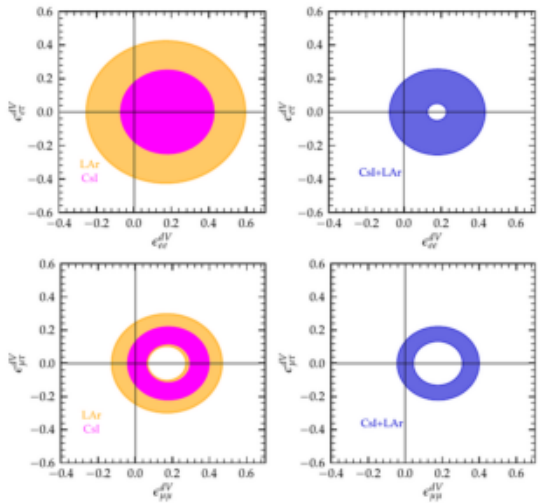


FIG. 5: 90% C.L. (2 d.o.f.) allowed regions assuming one nonuniversal and one flavor-changing neutrino NSI, extracted from the analysis of CsI (magenta), LAr (orange) and the combined CsI + LAr (blue) data. The upper panel shows the result in the $(\epsilon_{ee}^{dV}, \epsilon_{e\tau}^{dV})$ plane while the lower panel shows the corresponding result in $(\epsilon_{\mu\mu}^{dV}, \epsilon_{\mu\tau}^{dV})$. The analysis of CsI data includes only CEνNS interactions.

4.3. Neutrino NGI

We now turn our attention on a more general description of exotic new physics, namely NGI that go beyond the typical vector NSI that arise in gauge extensions of the SM. In this context, we aim to explore additional types of Lorentz-invariant interactions involving also scalar and tensor terms. In full generality, all types of NGI could in principle exist at the same time. However, for simplicity, in our analysis we only allow up to two new interactions to be present at the same time (while setting the third one to zero), assuming universal couplings (i.e. $C_X^u = C_X^d \equiv C_X^q$) and neglecting the ES data in the analysis.

Figure 7 shows the $\Delta\chi^2$ profiles for a single NGI (upper row) given in terms of the C_X^q, and the 90% C.L. (2 d.o.f.) allowed regions for neutrino NGI in the (C_X^q, C_Y^q) plane (lower row), with $C_X^q = g_{\nu X} \cdot g_{qX}$ and $X = S, V, T$, see Eqs. (9), (10) and (11). The results show that, although a

CEvNS-pyEFT native approximate reproduction

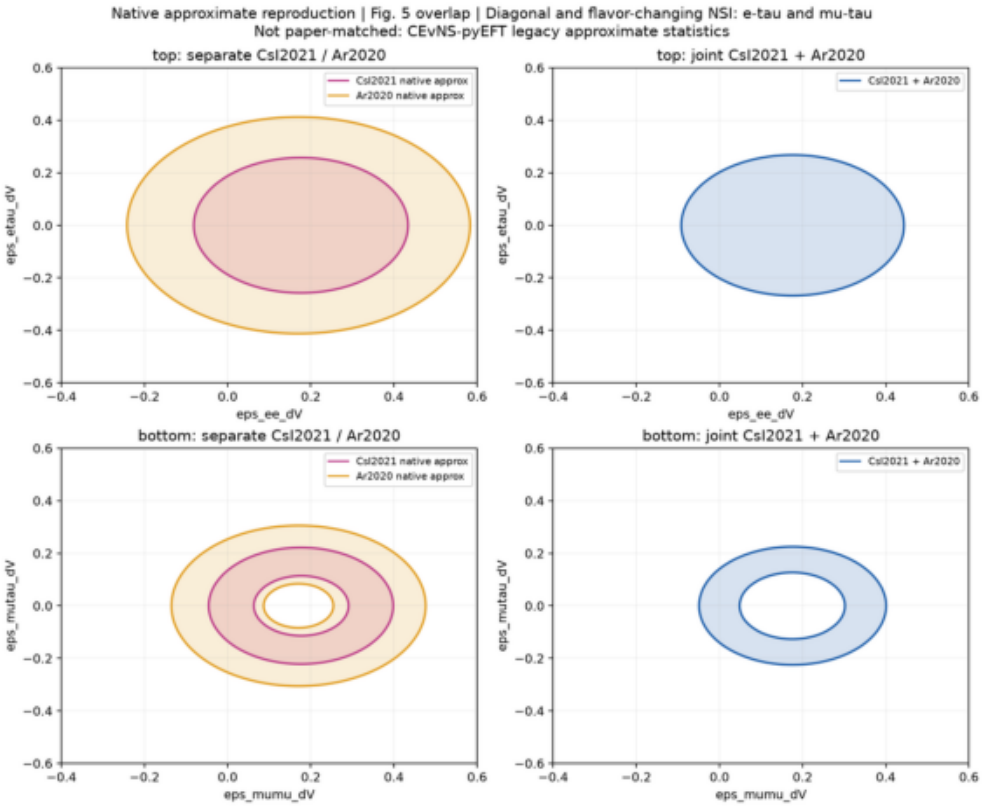


Fig06 native approx comparison

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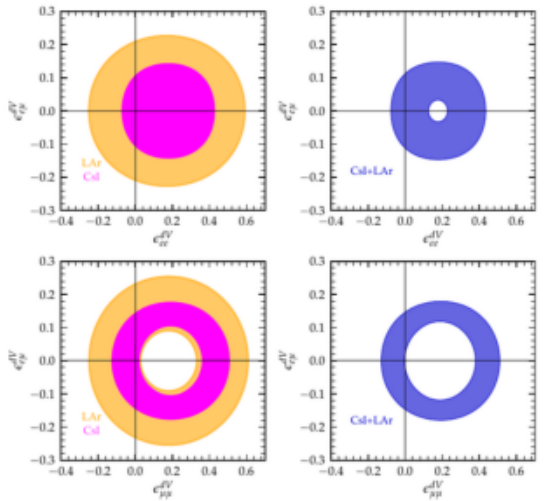


FIG. 6: Same as Fig. 5 for the planes $(\epsilon_{ee}^{dV}, \epsilon_{e\mu}^{dV})$ and $(\epsilon_{\mu\mu}^{dV}, \epsilon_{e\mu}^{dV})$, in the upper and lower panel, respectively.

new scalar or tensor interaction actually improves the fit of LAr data alone, such a new NGI does not fit the CsI data set, so that the combined analysis prefers a SM explanation. For the scenario with a new vector interaction instead, both LAr and CsI lead to two χ^2 minima, one of them at $C_V^q = 0$ and a second one at $C_V^q \sim 0.18$. In the region between the two minima, the extra vector contribution cancels the SM one, worsening the fit. If two interactions are simultaneously present, see lower panels, one can also expect interferences between the NGI and the SM interactions, as well as between different NGI. For the scalar versus tensor (S-T) case, the allowed region, indicated in the left panel, is centered around the SM solution, with $C_T^q = C_S^q = 0$, with no other solutions allowed at the same confidence level. In the other two cases, however, degenerate solutions with similar goodness of fit as the SM solution appear. In particular, in the scalar versus vector (S-V) case, shown in the middle panel, the combination of the two data sets reduces the ring-shaped contour obtained with LAr data to two smaller disjoint regions, one centered around the SM and a second one around $C_V^q \sim 0.18$, as already seen for the one-dimensional profiles. Similarly, for the vector versus tensor (V-T) case, the combined analysis of LAr and CsI data results in a ring-

CEvNS-pyEFT native approximate reproduction

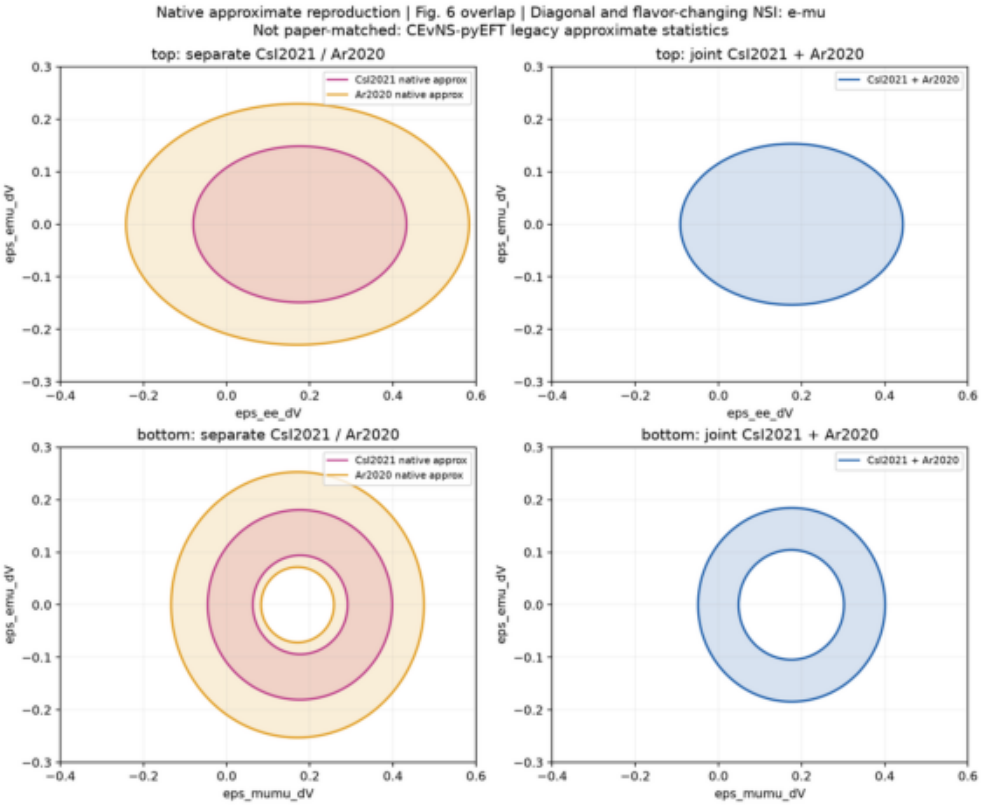


Fig07 NGI not comparable

Fig. 7 为 NGI scalar/vector/tensor benchmark；当前结果明确标记为 NGI_MISSING / NOT_COMPARABLE，没有使用 NSI 图替代。

Paper original page render: Fig. 7

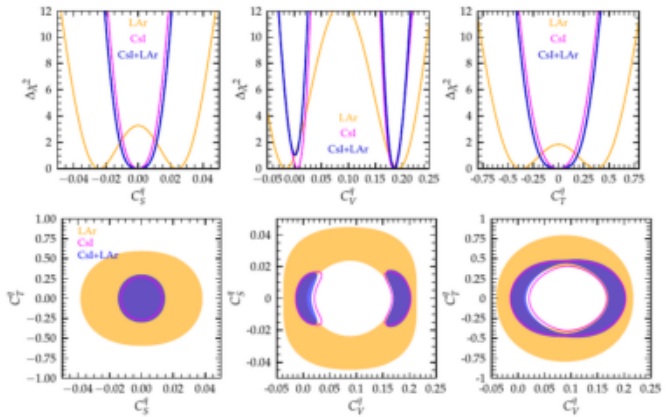


FIG. 7: Upper panels: $\Delta\chi^2$ profiles for single NGI from the CsI (magenta), LAr (orange) and CsI + LAr (blue) analyses. The analysis of CsI data includes only CEνNS interactions. Lower panels: 90% C.L. (2 d.o.f.) allowed regions on neutrino NGI allowing two NGI couplings at a time.

shaped allowed region including the SM solution $C_V^q = C_T^q = 0$. In all cases, the addition of the CsI data substantially improves the sensitivity to these new interactions and is competitive to existing results [25], extracted recently from the analysis of Dresden-II data [10]. Furthermore, one sees a tiny part of the parameter space where the combined contour extends outside the CsI allowed region. This can be understood from the LAr and CsI χ^2 profiles in the upper panels of Fig. 7, which show how in some cases the best fit values for the couplings in the LAr analysis lie outside the region preferred by the CsI data.

The 1σ limits obtained from the combined CsI+LAr analysis on vector, scalar and tensor NGI parameters (setting all the others to zero) read

$$\begin{aligned} C_V^q &= [-0.003, 0.003] \cup [0.18, 0.19], \\ C_S^q &= [-0.009, 0.009], \\ C_T^q &= [-0.19, 0.19]. \end{aligned} \tag{45}$$

No native reproduction: NGI scalar/tensor not implemented in this comparison

Fig. 7 is NGI scalar/vector/tensor, not vector NSI epsilon.

This run only covers the NSI overlap of Fig. 3-6 with current CEvNS-pyEFT Flavor machinery.

Status: NGI_MISSING / NOT_COMPARABLE.

No fake scalar/tensor arrays were generated.